Unbiased Estimators & Confidence Intervals

Section 1: Unbiased Estimator of Population Mean & Variance

1 Given sample observations $x_1, \ldots, x_n$, what is the formula for the unbiased estimator s_{n-1}^2 of the population variance σ^2?

a. $s_{n-1}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$

b. $s_n^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$

c. $s^2 = \frac{1}{n+1}\sum_{i=1}^{n} x_i^2$

d. $s^2 = \frac{(\sum x_i)^2}{n-1}$

2 A sample of $n = 5$ values has $\sum x_i = 30$ and $\sum x_i^2 = 210$. Calculate the unbiased estimate of the population variance s_{n-1}^2.

a. 7.5

b. 6.0

c. 30

d. 9.0

3 If a sample standard deviation computed with denominator n is $s_n = 4.0$ for $n = 16$, what is the unbiased standard deviation s_{n-1}?

a. $s_{n-1} = 4.0 \times \sqrt{\frac{16}{15}} \approx 4.13$

b. $s_{n-1} = 4.0 \times \sqrt{\frac{15}{16}} \approx 3.87$

c. $s_{n-1} = 4.0$

d. $s_{n-1} = 4.0 \times \frac{16}{15} \approx 4.27$

Section 2: Confidence Intervals for Mean (Known Variance)

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4 A random sample of $n = 64$ items from a population with known $\sigma = 12$ has mean $\bar{x} = 50$. What is the 95% confidence interval for μ ($z^* = 1.96$)?

a. $[47.06, 52.94]$

b. $[48.50, 51.50]$

c. $[44.12, 55.88]$

d. $[49.63, 50.37]$

5 To reduce the margin of error of a confidence interval by a factor of 3, the sample size must be multiplied by:

a. 9

b. 3

c. $\sqrt{3}$

d. 6

6 What is the critical value z^* for a 99% confidence interval?

a. 2.576

b. 1.960

c. 1.645

d. 2.326

Section 3: Confidence Intervals for Mean (Unknown Variance, t-distribution)

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7 When constructing a confidence interval for μ with unknown population variance σ^2 from a normal sample of size $n = 10$, which distribution is used?

a. Student's t-distribution with 9 degrees of freedom

b. Standard Normal Z-distribution

c. Student's t-distribution with 10 degrees of freedom

d. Chi-square distribution with 9 degrees of freedom

8 How does the critical value t^* compare to z^* for the same confidence level and finite sample size?

a. $t^* > z^*$, because the t-distribution has heavier tails to reflect estimation uncertainty

b. $t^* < z^*$, because t is more compact

c. $t^* = z^*$ always

d. t^* can be smaller or larger depending on $\bar{x}$

9 A sample of $n = 16$ has $\bar{x} = 22.0$ and $s_{n-1} = 2.0$. Given $t_{0.025,\,15} = 2.131$, find the 95% confidence interval for μ.

a. $[20.93, 23.07]$

b. $[21.47, 22.53]$

c. $[19.87, 24.13]$

d. $[20.00, 24.00]$

Section 4: Interpretation of Confidence Levels

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10 Which statement correctly conveys the meaning of a '95% confidence level'?

a. If many random samples are drawn and intervals constructed, 95% of them will contain the true population parameter

b. There is a 95% probability that the true parameter lies inside the specific interval $[47, 53]$

c. 95% of sample data values lie between 47 and 53

d. The sample mean has a 95% chance of being correct

11 If a 95% confidence interval for μ is $[102, 118]$, what is the conclusion for the two-tailed test $H_0 : \mu = 100$ vs $H_1 : \mu \neq 100$ at $\alpha = 0.05$?

a. Reject H_0 at the 5% level, because 100 lies outside the 95% confidence interval

b. Fail to reject H_0, because 100 is close to 102

c. Accept H_0 as proven

d. The test is inconclusive without a P-value

12 What happens to the width of a confidence interval if the confidence level is increased from 90% to 99% (with sample size fixed)?

a. The interval widens

b. The interval narrows

c. The interval width remains identical

d. The interval shifts to the right

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